FLUMAZENIL

Seizure risk — not routine in overdose

Short-acting

CHEMICAL NAME Ethyl 8-fluoro-5,6-dihydro-5-methyl-6-oxo-4H-imidazo[1,5-a][1,4]benzodiazepine-3-carboxylate

CLASS Benzodiazepine receptor antagonist (antidote)

TRADE NAMES Anexate · Romazicon · Flumazenil injection

THERAPEUTIC USES / INDICATIONS

- Reversal of benzodiazepine sedation after procedures (e.g. endoscopy with midazolam) or anaesthesia.
- Iatrogenic benzodiazepine overdose in non-dependent patients.
- Not recommended for routine use in mixed/unknown overdoses.

ADMINISTRATION

- **IV** slowly; titrate in small increments.
- Patient monitoring for re-sedation.

DOSAGES

- **Adult:** 200 mcg IV over 15 s, then 100 mcg every 60 s as needed; usual total 300–600 mcg; max 1 mg (2 mg in ICU).
- **Children:** 10 mcg/kg (max 200 mcg), repeat every minute (max 50 mcg/kg or 1 mg).
- **Re-sedation:** infusion 100–400 mcg/h.

PHARMACOKINETICS

- **Onset:** 1–2 min. **Duration:** 30–60 min (shorter than most benzodiazepines).
- Metabolised in liver. **t½:** ~1 h.

PHARMACODYNAMICS

- Competitive antagonist at the **benzodiazepine binding site of GABA-A receptors** → reverses sedation, respiratory depression (partially), amnesia.
- Also reverses Z-drugs (zolpidem).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Anxiety, agitation, dizziness.
- Flushing.

ADVERSE EFFECTS (SERIOUS)

- Seizures (benzodiazepine-dependent patients, epilepsy, co-ingestion of TCAs or other proconvulsants).
- Acute benzodiazepine withdrawal.
- Arrhythmias (with TCA overdose); re-sedation.

CONTRAINDICATIONS & CAUTIONS

- Patients on benzodiazepines for life-threatening conditions (status epilepticus, raised ICP).
- Mixed overdose with TCAs or proconvulsant drugs; QRS widening.
- Benzodiazepine dependence (relative).

KEY INTERACTIONS

- Reverses benzodiazepines and Z-drugs only (not opioids, alcohol, barbiturates).

NURSING MANAGEMENT

- Use in controlled settings (procedural reversal).
- Monitor RR, SpO_2, GCS for at least 2 h (re-sedation).
- Seizure precautions; have IV benzodiazepine available.
- Check ECG in overdose (TCA).

PATIENT EDUCATION

- You may feel anxious or nauseated briefly.
- Do not drive or sign documents for 24 h after sedation.

KEY POINTS

- Benzodiazepine antagonist.
- Seizure risk; avoid in dependence or TCA co-ingestion.
- 200 mcg IV, then 100 mcg/min (max 1 mg).
- Short-acting — watch for re-sedation.

SOURCES CHECKED **[989]** Flumazenil 100 micrograms/ml solution for injection/infusion (UK SmPC) **[990]** Flumazenil, IV (US label) **[991]** Flumazenil, IV (US label, Fresenius Kabi USA)

Full references p. 432–457